Open-Canopy: Towards Very High Resolution Forest Monitoring

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Motivation

Canopy Height:

- Related to biomass / carbon cycle
- Measured with terrestrial or aerial LiDAR: 50M$ / 5 years for France

Solution:

- Estimate canopy height from cheap and frequent satellite images
- Leverage paired images / height maps

Open-Canopy:

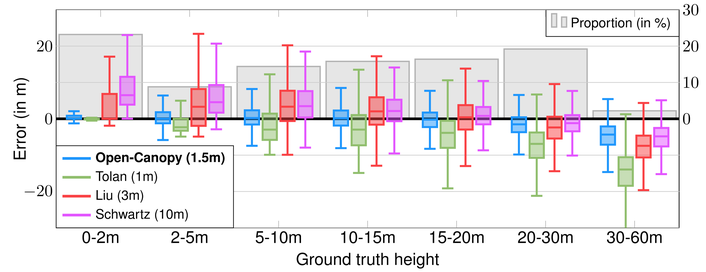
- First open-access, country-scale benchmark at very high-resolution (1.5 m)
- 87,000 km^2 across France
- Spot6/7 RGB+nIR pansharpened 1.5 m
- Aerial LiDAR ground truth
- Clean train/val/test
- Extensive benchmark of SOTA models

Existing datasets

	Dataset	access: code	access: img	access: GT	extent: scope	extent: surface $\times 10^3$ km^2	images: sensor	images: res. in m	height ground truth: sensor	height ground truth: res. in m	height ground truth: samples $\times 10^6$
HR	Schwartz	✗	?	⚙ ?	France	588	S1/S2	10	GEDI	25	90
HR	Lang	✔	⬇	⬇	Global	14k	S2	10	GEDI	25	600
HR	Potapov	✗	?	⚙ ?	Global	150k	Landsat	30	GEDI	25	372
HR	Pauls	✔	?	⚙ ?	Global	2621	S1/S2	10	GEDI	25	?
VHR	Tolan	✗	$	⬇	US	5.8	MAXAR	1.2	ALS+GEDI	1	5800
VHR	Wagner	✗	🔒 ?	?	US	3.8	NAIP	0.6	ALS	1	3784
VHR	Liu	✗	🔒 ?	?	Europe	700	Planet	3	ALS	3	77,777
	Open-Canopy	✔	⬇	⬇	France	87	SPOT 6-7	1.5	ALS	1.5	38,876

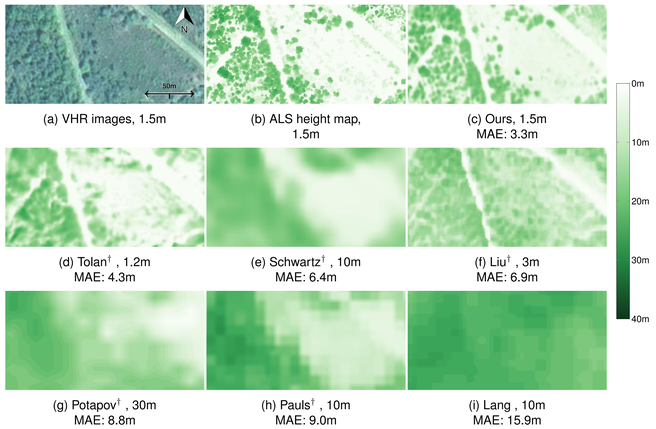
⬇ direct download · $ commercial · ⚙ complex pre-processing · 🔒 special access · ? no train test split · S1 Sentinel-1 · S2 Sentinel-2

Error Distribution



- Evaluated on OpenCanopy test set
- Higher canopies are harder to estimate

Comparison to Canopy Height Products

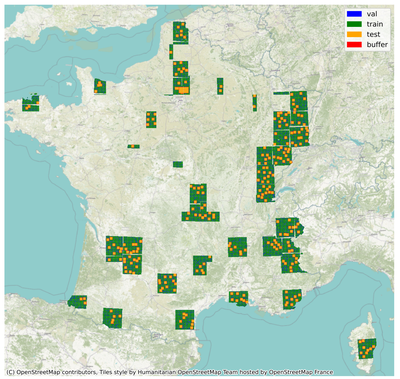


(a) VHR images, 1.5m · (b) ALS height map, 1.5m · (c) Ours, 1.5m MAE: 3.3m

(d) Tolan† , 1.2m MAE: 4.3m · (e) Schwartz† , 10m MAE: 6.4m · (f) Liu† , 3m MAE: 6.9m

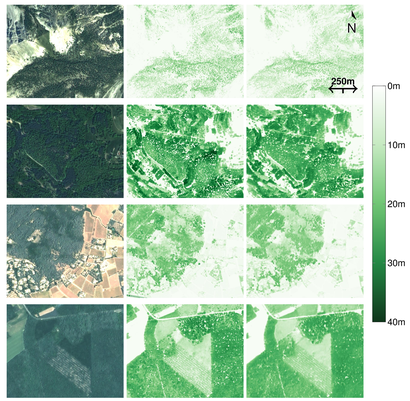
(g) Potapov† , 30m MAE: 8.8m · (h) Pauls† , 10m MAE: 9.0m · (i) Lang , 10m MAE: 15.9m

Open Canopy



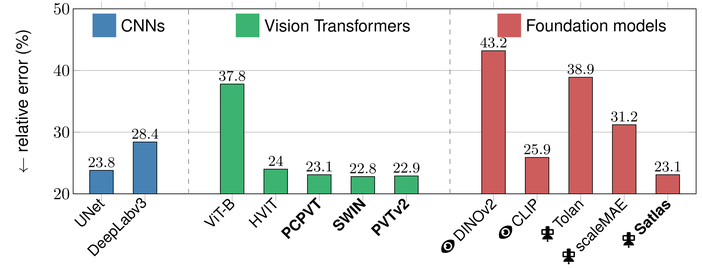
(a) Splits

Qualitative Results

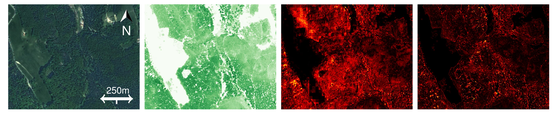


VHR image · ALS height map · PVTv2 height map

Benchmark



- Hierarchical ViTs perform best
- LVMs underperform



(a) VHR image · (b) ALS-derived height · (c) Vit-B Error · (d) PVTv2 Error

OOD Evaluation: USA

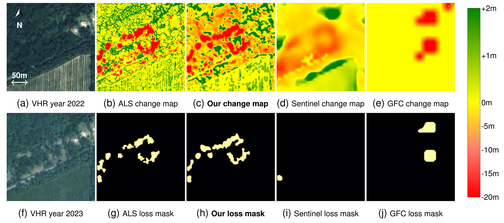
	Input data	Training area	MAE in m	nMAE in %	RMSE in m	Bias in m	Tree cov. IoU in %
Tolan Map	MAXAR	US	**2.02**	47.4	3.58	**0.57**	**70.5**
PVTv2	NAIP	OpenCanopy	4.38	71.0	6.42	2.78	49.6
PVTv2	SPOT 6-7	OpenCanopy	2.08	**33.9**	**3.20**	0.90	61.8

- **Different sensor (MAXAR):** fails to generalize
- **Same sensor (SPOT 6/7):** performs on par with a product pretrained on US data!

Open Canopy-Δ: Benchmarking Change Detection

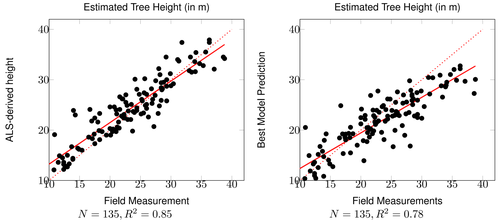
- **Study Area:** 2 consecutive ALS for 16k ha
- 73 changes validated by experts

	Precision (%)	Recall (%)	IoU (%)
Schwartz	**63.5**	3.2	3.1
GFC	0.9	11.1	0.8
PVTv2 (ours)	53.8	**54.3**	**37.0**



(a) VHR year 2022 · (b) ALS change map · (c) **Our change map** · (d) Sentinel change map · (e) GFC change map

(f) VHR year 2023 · (g) ALS loss mask · (h) **Our loss mask** · (i) Sentinel loss mask · (j) GFC loss mask

Field Verification

- 135 plots with a 15 m radius
- National Forest Inventory (INF)
- **ALS heights:** align closely with field measurements
- **Model predictions:** strongly correlated to field measurements